

SCAS: A framework for Detecting Structural Anomalies in European Energy Generation Networks with Event Validation (2016-2024)

Bruno Priantti

May 2026

Abstract

Detecting structural changes in interconnected energy systems can help reveal how external events affect energy generation patterns. We propose the Structural Change Alert Signal (SCAS), a network-based framework for identifying periods of potential structural change in European energy generation. The analysis uses monthly data from ten countries and five energy categories, applying log-differencing and standardization before constructing sliding-window correlation networks. Four network metrics are calculated for each window: density, clustering, modularity, and assortativity. Matrix Profile is then applied to these metrics to identify anomalous temporal patterns, which are combined into a binary SCAS. Across 102 monthly windows from 2016 to 2024, the SCAS identifies periods coinciding with 24 of 29 documented events (82.8%) within a two-month tolerance window, with most matches involving density, clustering, and modularity. The results demonstrate the potential of SCAS as an exploratory framework for monitoring structural changes in interconnected energy generation networks. The implementation is available at: <https://github.com/quantyma/quantyma-research/tree/main/quantyma-article-005>

Keywords: Energy Generation, Correlation Networks, Complex Networks, Structural Anomalies, Matrix Profile, Time Series, European Energy Systems.

1 Introduction

The integration of interconnected power systems across national borders has become a central feature of the European energy landscape [Trebbien et al., 2024]. However, the structural relationships between generation profiles of different countries are not static; they evolve in response to geopolitical disruptions, market shocks, and climate anomalies [Rydin Gorjão et al., 2022]. Understanding when and how these relationships change is important for assessing the evolution and resilience of interconnected energy systems.

Traditional correlation-based analyses of energy systems often assume stationary relationships over time [Baek et al., 2025]. This assumption can obscure structural changes that may precede or accompany major disruptions, such as the 2022 European energy crisis [Trebbien et al., 2024]. While recent work has applied network-based methods to energy systems [Han et al., 2022], systematic detection of anomalous changes in correlation network structure remains underexplored.

This paper presents the Structural Change Alert Signal (SCAS), a network-based framework for identifying periods associated with potential structural changes in European energy generation networks. The framework integrates monthly generation data from ten countries across five energy categories, applies first-order differencing and normalization to reduce trend and scale effects, and constructs sliding-window correlation networks. Four network metrics, namely density, clustering, modularity, and assortativity, are computed for each temporal window. Matrix Profile is then applied to the resulting metric time series to identify anomalous temporal patterns [Yeh et al., 2016]. These anomalies are combined into the binary SCAS, which marks periods in which at least one network metric exhibits an anomalous pattern.

The SCAS is evaluated against 29 documented events spanning six categories: geopolitical, market, climate, infrastructure, policy, and sanitary. The validation measures the temporal coincidence between SCAS alerts and documented events using a two-month tolerance window. The resulting framework provides a reproducible approach for monitoring structural changes in interconnected energy generation networks and examining their temporal relationship with external events.

2 Methodology

2.1 Data Collection and Preprocessing

This study uses the Real-E benchmark, constructed from the ENTSO-E Transparency Platform and covering 39 European countries [Shao et al., 2025]. We adopt the O20 version, which filters variables with an overlapping rate below 20%, yielding a subset with improved temporal coverage.

We use the “Actual Generation per Production Type” subset at the national level (CTY), consolidating the original production types into five broader categories: Fossil, Nuclear, Hydro, Wind, and Solar. This aggregation reduces dimensionality while retaining broad distinctions among generation technologies.

From this pool, we select ten major, highly interconnected countries in the European electricity market: Austria, Belgium, Czech Republic, France, Germany, Italy, Netherlands, Poland, Spain, and Switzerland [ENTSO-E, 2025].

The original 15-minute generation data are aggregated into monthly totals. The resulting country-level series are then merged using the monthly time index (MTU), with variables renamed according to their country code and generation category. The resulting unified multivariate dataset contains 50 time series, corresponding to ten countries across five generation categories.

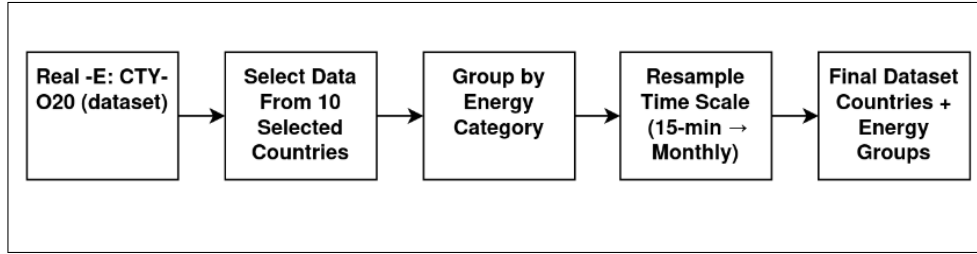


Figure 1: Data Collection and Preprocessing Pipeline

2.2 Time Series Preprocessing

The raw generation data exhibit two characteristics that can distort correlation-based analysis: long-term trends and large differences in magnitude across countries and generation categories. To reduce these effects, we apply two sequential transformations to each of the 50 time series: logarithmic differencing and z-score normalization.

First, a $\log 1p$ transformation is applied to each series. For a generation series X_t , the transformed value is defined as

$$Y_t = \log(1 + X_t). \quad (1)$$

The logarithmic transformation reduces the influence of large values and compresses differences in scale associated with the magnitude of electricity generation. First-order differencing is then applied to the transformed series:

$$\Delta Y_t = Y_t - Y_{t-1} = \log(1 + X_t) - \log(1 + X_{t-1}). \quad (2)$$

This transformation converts the original generation levels into month-to-month log changes, reducing the influence of long-term trends while retaining short-term temporal variation.

Finally, each differenced series is standardized independently using z-score normalization:

$$Z_t = \frac{\Delta Y_t - \mu_{\Delta Y}}{\sigma_{\Delta Y}}, \quad (3)$$

where $\mu_{\Delta Y}$ and $\sigma_{\Delta Y}$ denote the mean and standard deviation of the corresponding differenced series. This step places all 50 time series on a common scale, preventing differences in magnitude from dominating the subsequent correlation analysis. The resulting standardized series are used in all subsequent network construction and anomaly detection steps.

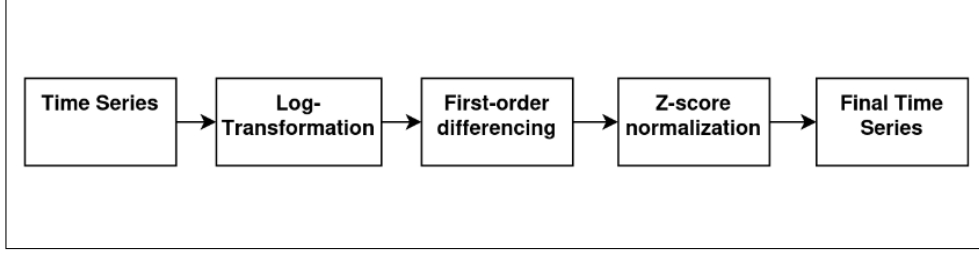


Figure 2: Time Series Preprocessing

2.3 Temporal Correlation Network Construction

To characterize the temporal evolution of the relationships among the 50 country–category generation series, we construct a sequence of time-varying correlation networks. The procedure transforms the multidimensional time series into overlapping temporal windows, computes the pairwise correlation structure within each window, and represents the resulting relationships as weighted signed graphs. This process is illustrated in Figure 3.

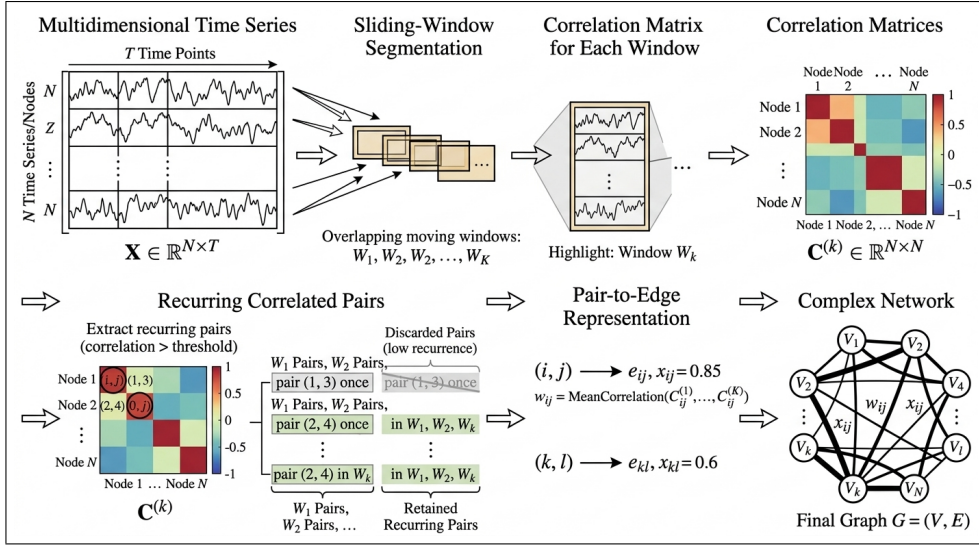


Figure 3: Temporal Correlation Network Construction

Let $\mathbf{X} * t = [X * 1, t, X_{2,t}, \dots, X_{50,t}]$ denote the vector containing the 50 preprocessed time series at month t . To characterize the evolution of their pairwise relationships over time, we employ a rolling-window approach with a window length of $L = 12$ months and a step size of one month. Given the 113 monthly observations from February 2015 to June 2024, this procedure produces

$$W = T - L + 1 = 113 - 12 + 1 = 102 \quad (4)$$

overlapping temporal windows. The w -th window is defined as

$$\mathcal{W}w = \mathbf{X}w, \mathbf{X}w + 1, \dots, \mathbf{X}w + L - 1, \quad w = 1, \dots, 102. \quad (5)$$

For each window, pairwise Pearson correlations are computed among the available time series with non-zero variance within that window. The resulting correlations form a symmetric matrix

$$\mathbf{R}^{(w)} = \left[r_{ij}^{(w)} \right] \in [-1, 1]^{50 \times 50}, \quad (6)$$

where $r_{ij}^{(w)}$ denotes the Pearson correlation between series i and j within window w . Since the correlation matrix is symmetric, only the upper triangular elements are considered, yielding unique pairwise correlations per window.

$$N_{\text{pairs}} = \binom{50}{2} = \frac{50(50-1)}{2} = 1225 \quad (7)$$

Each correlation matrix is subsequently transformed into a weighted signed graph

$$G^{(w)} = (V, E^{(w)}, \mathbf{W}^{(w)}), \quad (8)$$

where the node set V contains the country–category pairs and an edge (i, j) is included when

$$(i, j) \in E^{(w)} \iff |r_{ij}^{(w)}| \geq \tau, \quad \tau = 0.5. \quad (9)$$

The weight associated with each retained edge is the original correlation,

$$w_{ij}^{(w)} = r_{ij}^{(w)}. \quad (10)$$

Consequently, positive edge weights represent positive co-movement between two country–category series, whereas negative weights represent opposing temporal variation. The resulting sequence of graphs, represents the temporal evolution of the correlation structure among the 50 country–category variables.

For computational efficiency, each graph is represented as an edge list containing the connected node pairs and their corresponding weights. For each temporal window, the analysis retains the window end date, the full correlation matrix $\mathbf{R}^{(w)}$, the upper-triangular mask used to identify the 1,225 unique pairs, and the resulting edge list. This representation preserves the information required for subsequent network metric computation and temporal anomaly detection while avoiding redundant storage of symmetric graph connections.

$$\mathcal{G} = \{G^{(1)}, G^{(2)}, \dots, G^{(102)}\}. \quad (11)$$

2.4 Network Metrics

For each temporal network, we compute four complementary network metrics that characterize global connectivity, local cohesion, community organization, and hub topology. These metrics provide a multi-dimensional representation of the structural evolution of the network across the rolling windows.

Density measures the fraction of observed edges relative to all possible edges, reflecting the overall connectivity of the network [Wasserman and Faust, 1994]. Higher values indicate a more densely connected structure, while changes over time may reveal variations in the overall strength of network relationships.

Clustering coefficient quantifies the tendency of neighboring nodes to be connected to each other, capturing local cohesion [Watts and Strogatz, 1998]. Changes in this metric indicate variations in the formation of tightly interconnected groups, which may occur even when global network density remains relatively stable.

Modularity measures the strength of the division of the network into communities [Newman and Girvan, 2004]. Higher values indicate more clearly separated communities, allowing temporal variations to capture changes in the organization or separation of groups within the network.

Degree assortativity measures the correlation between the degrees of connected nodes [Newman, 2002]. Positive values indicate that high-degree nodes tend to connect with other high-degree nodes, whereas negative values indicate disassortative mixing. This metric therefore captures changes in how highly connected nodes are organized within the network.

Together, these four metrics provide complementary information about the structural evolution of the temporal networks. Their resulting time series are subsequently analyzed for anomalous configurations using the Matrix Profile approach.

Table 1: Network metrics used in this study.

Metric	What it measures	Reference
Density	Overall connectivity of the network	Wasserman and Faust [1994]
Clustering coefficient	Local cohesion among neighboring nodes	Watts and Strogatz [1998]
Modularity	Strength of division into communities	Newman and Girvan [2004]
Degree assortativity	Tendency of hubs to connect to other hubs	Newman [2002]

2.5 Anomaly Detection via Matrix Profile

To detect anomalous regime shifts in the network metrics, we apply the Matrix Profile algorithm [Yeh et al., 2016] to each of the four metric time series. Matrix Profile computes, for each subsequence of a given length, the distance to its nearest neighbor across the entire series. Subsequences with large distances (discords) represent patterns that are maximally different from all others, effectively identifying structural anomalies.

For each metric, we use a subsequence length of 6 months and retain the five subsequences with the highest distances (top-5 discords) as candidate anomalies. This produces a total of 20 anomaly candidates across the four metrics, which are then validated against documented events in the following section.

2.6 Structural Change Alert Signal (SCAS)

Following anomaly detection, we construct a binary Structural Change Alert Signal (SCAS) to identify periods associated with potential structural changes in the network. Whenever an anomaly is detected in any of the four network metrics, the SCAS is activated, taking the value $S_t = 1$. The signal remains active for the subsequent two months, after which it returns to the OFF state, represented by $S_t = 0$, unless another anomaly is detected.

The SCAS does not indicate that a structural change has necessarily occurred. Instead, it defines a two-month alert window following an anomalous observation, during which a structural change event may have occurred. This provides a consistent temporal criterion for comparing detected anomalies with documented external events.

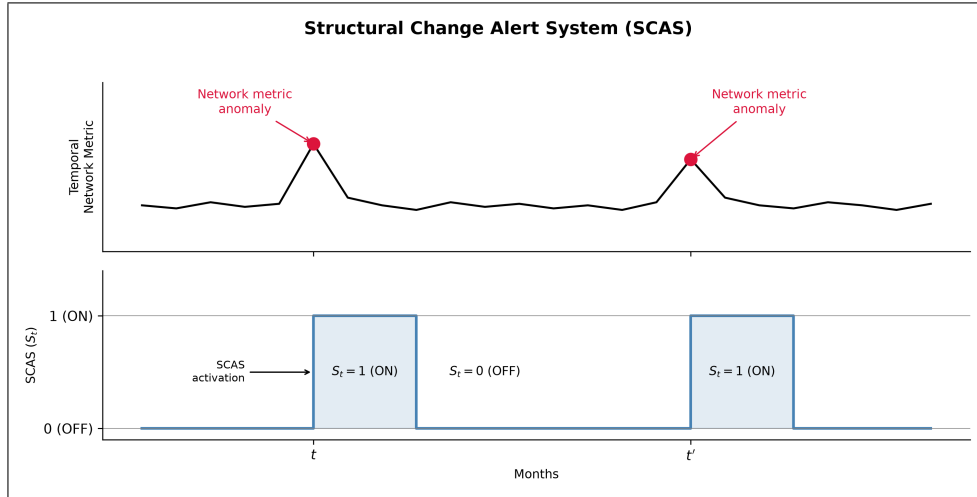


Figure 4: SCAS diagram

2.7 Validation against Documented Events

To assess whether the detected anomalies correspond to real-world disruptions, we validate them against 29 documented events spanning six categories: geopolitical, market, climate, infrastructure, policy, and sanitary. These events were compiled from public sources-including the ENTSO-E Transparency Platform, European Commission reports, and news archives-and cover major disruptions that affected the European energy system between 2016 and 2024. Table 2 lists all events used in the validation.

An anomaly is considered a match if it falls within a two-month tolerance of a documented event. This tolerance accounts for the fact that both the rolling window and the Matrix Profile operate on monthly aggregates, introducing a natural lag between the occurrence of an event and its manifestation in the correlation structure.

For each match, we record the metric that detected the anomaly, the window date, the metric value, and the associated event category. We then compute the overall match rate (number of matched anomalies divided by total anomalies) and analyze the distribution of matches across metric types and event categories.

Table 2: Documented events used for validation, grouped by category.

Date	Event	Category
2016-06	Brexit	Geopolitical
2018-07	Drought and heatwave	Climate
2018-09	CO ₂ price crisis	Market
2019-01	German nuclear phase-out (1)	Infrastructure
2019-06	Record heatwave	Climate
2019-06	Clean Energy Package	Policy
2020-01	Brexit formalized	Geopolitical
2020-03	COVID-19	Sanitary
2020-04	Negative oil prices	Market
2020-09	55% renewable target	Policy
2021-01	Biden rejoins Paris Agreement	Geopolitical
2021-01	German nuclear phase-out (2)	Infrastructure
2021-07	Germany/Belgium floods	Climate
2021-09	Gas crisis	Market
2021-10	Global energy crisis	Market
2022-02	Ukraine invasion	Geopolitical
2022-04	French nuclear issues	Infrastructure
2022-05	REPowerEU	Policy
2022-06	Extreme drought	Climate
2022-08	Gas at €300/MWh	Market
2022-09	Nord Stream sabotage	Geopolitical
2022-12	Gas price cap	Market
2023-04	German nuclear phase-out (3)	Infrastructure
2023-06	French nuclear recovery	Infrastructure
2023-07	Hottest July on record	Climate
2023-10	Israel-Hamas war	Geopolitical
2023-10	Electricity market reform	Policy
2024-01	Mild winter	Climate

3 Results and Discussion

Time Series Preprocessing. The top panel shows the raw time series (in MW) for all analyzed variables, highlighting the heterogeneous scales and the presence of outliers and missing values across the 2015–2024 period. The bottom panel presents the standardized time series obtained via z-score normalization, where all variables are rescaled to zero mean and unit variance, enabling direct comparison and mitigating scale-related biases prior to model training.

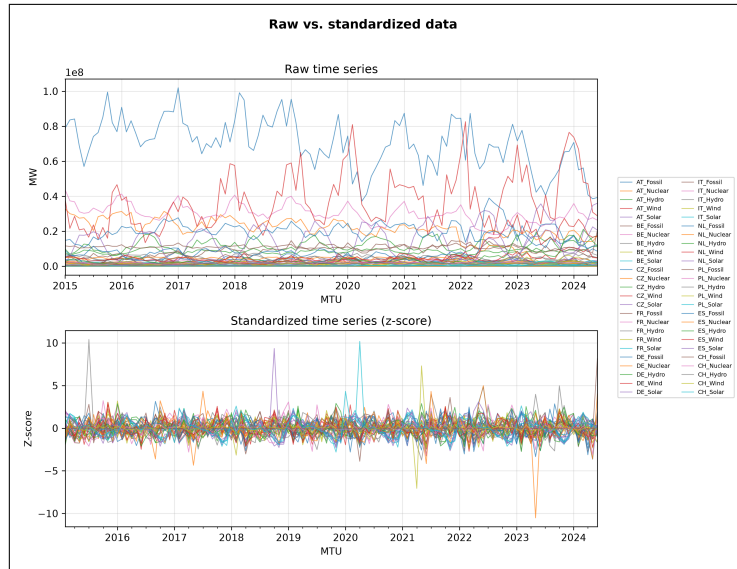


Figure 5: Time Series Preprocessing

Evolution of Network. The figure illustrates the temporal evolution of the energy generation correlation network from 2016-01 to 2024-01. Each panel corresponds to a snapshot of the signed weighted graph built from the 50 country–category generation series, where nodes represent country–category pairs and edges denote significant pairwise correlations ($|r| \geq 0.5$). Red edges indicate positive correlations (co-movement), while blue edges indicate negative correlations (opposing temporal variation). The progressive densification and shifting connectivity patterns across snapshots reveal structural changes in the interdependence of energy generation profiles over time.

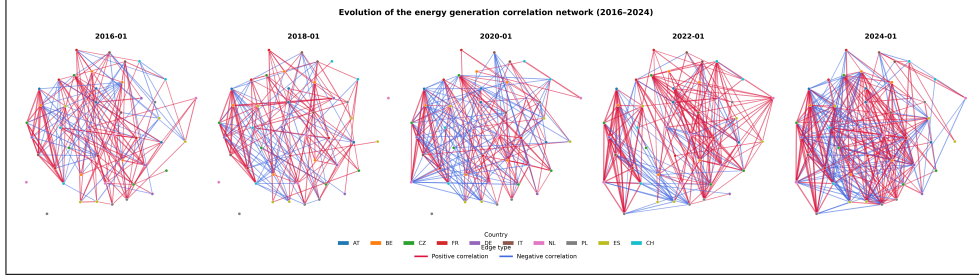


Figure 6: Evolution of Network

Temporal Network Metrics. The figure displays the monthly evolution of four global network metrics: density, clustering, modularity, and assortativity computed on the rolling-window correlation graphs over the 2016–2024 period. Each metric is shown in its own panel, with the corresponding time series plotted in blue. Red markers indicate months flagged as anomalous by the anomaly detection procedure, i.e., observations whose metric values deviate significantly from the local temporal baseline. The bottom panel reports the binary anomaly flag (0/1), where shaded red bars highlight the months identified as anomalous across the analyzed metrics. The co-occurrence of anomalies across multiple metrics (e.g., around 2018, 2020, and 2021–2022) suggests periods of structural reconfiguration in the energy generation correlation network, potentially associated with major market, geopolitical, or climate events.

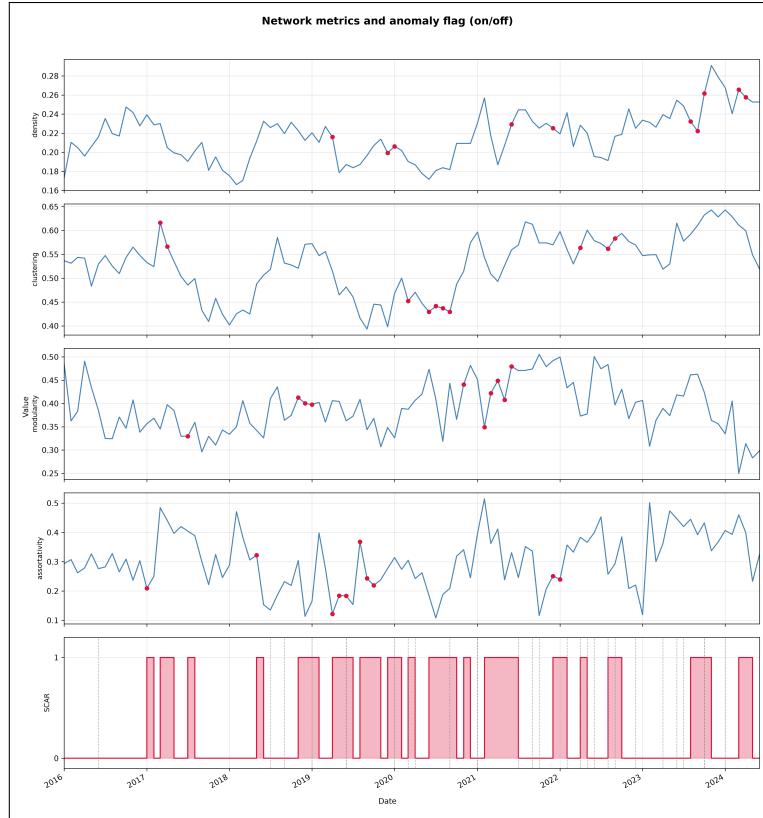


Figure 7: Temporal Network Metrics

To examine the temporal relationship between documented external events and structural changes in the network, we compare the event chronology with the detected discords using a 2-month tolerance window. The resulting correspondences are presented in Table 3, together with the event category, number and dates of matched discords, and the network metrics involved.

Table 3: Alignment between documented events and detected discords.

event_date	event	category	matched
2016-06	Brexit	geopolitics	False
2018-07	Drought and Heatwave	climate	True
2018-09	CO2 Crisis	market	True
2019-01	German Nuclear Phase-out (1)	infrastructure	True
2019-06	Record Heatwave	climate	True
2019-06	Clean Energy Package	policy	True
2020-01	Brexit Formalized	geopolitics	True
2020-03	COVID-19	health	True
2020-04	Negative Oil Prices	market	True
2020-09	55% Renewable Energy Target	policy	True
2021-01	Biden Rejoins the Paris Agreement	geopolitics	True
2021-01	German Nuclear Phase-out (2)	infrastructure	True
2021-07	Germany/Belgium Floods	climate	True
2021-09	Gas Crisis	market	False
2021-10	Global Energy Crisis	market	True
2022-02	Invasion of Ukraine	geopolitics	True
2022-04	French Nuclear Power Issues	infrastructure	True
2022-05	REPowerEU	policy	True
2022-06	Extreme Drought	climate	True
2022-08	Gas Prices Reach €300/MWh	market	True
2022-09	Nord Stream Sabotage	geopolitics	True
2022-12	Gas Price Cap	market	False
2023-04	German Nuclear Phase-out (3)	infrastructure	False
2023-06	French Nuclear Recovery	infrastructure	True
2023-07	Hottest July on Record	climate	True
2023-10	Israel-Hamas War	geopolitics	True
2023-10	Electricity Market Reform	policy	True
2024-01	Mild Winter	climate	True

4 Discussion

The results indicate that the temporal correlation network exhibits structural reconfigurations in the vicinity of documented exogenous events. Using a ± 2 -month tolerance window, 24 of 29 events (82.8%) are matched with at least one anomalous month. However, comparison with a random baseline shows that this match rate is only moderately above chance: random event dates yield 76.8% $\pm 6.7\%$, corresponding to a gain of 6.0 percentage points and a z -score of 0.90, below the conventional 5% significance threshold.

This result is partly explained by the high temporal density of anomalies and the relatively wide tolerance window, which make random matches likely. In addition, the limited number of documented events ($N = 29$) reduces statistical power, while both events and anomalies tend to cluster during periods of systemic stress.

Despite the limited statistical evidence, the matched events exhibit consistent structural patterns. Density, clustering, and modularity are the most frequently affected metrics and often co-occur around major episodes such as the 2019 record heatwave, the 2021 global energy crisis, and the 2022 invasion of Ukraine. Conversely, several documented events produce no detectable network anomaly, suggesting either limited propagation to the aggregate correlation structure or reduced sensitivity of the selected metrics.

Table 4: Match rate against random baseline (tolerance window of ± 2 months).

Method	Match Rate
Event-based matching	82.8% (24/29)
Random baseline (mean)	$76.8\% \pm 6.7\%$
Gain	+6.0 pp
z -score	0.90

5 Conclusion

This study presented the Structural Change Alert Signal (SCAS), a network-based framework for identifying periods of potential structural change in European energy generation networks. By combining temporal correlation networks, network-level metrics, and Matrix Profile anomaly detection, the framework provides a compact signal for monitoring changes in the structure of relationships among energy generation profiles.

Applied to data from ten European countries between 2016 and 2024, SCAS identified periods coinciding with 24 of 29 documented events within a two-month tolerance window. The results indicate that network density, clustering, and modularity were particularly relevant to the detected changes. While the event validation provides evidence of temporal coincidence, the comparison with a random baseline indicates that these results should be interpreted as exploratory rather than as evidence of a statistically established event–network relationship.

Future work can extend SCAS through sensitivity analyses of the network construction and anomaly detection parameters, larger event datasets, and additional network descriptors. These extensions may help assess the robustness and general applicability of the framework to other interconnected energy systems.

References

- Julius Trebbien, Anton Tausendfreund, Leonardo Rydin Gorjão, and Dirk Witthaut. Patterns and correlations in european electricity pricing. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 2024. doi: 10.1063/5.0201734.
- Leonardo Rydin Gorjão et al. Complexity and persistence of price time series of the european electricity spot market. *PRX Energy*, 1:013002, 2022.
- Changryong Baek et al. Detecting changes in correlation networks with application to functional connectivity of fmri data. *Psychometrika*, 2025.
- C Han et al. Complexity and persistence of price time series of the european electricity spot market. *PRX Energy*, 1:013002, 2022.
- Chin-Chia Michael Yeh et al. Matrix profile i: All pairs similarity joins for time series: A unifying view that includes motifs, discords and shapelets. *IEEE ICDM*, 2016.
- Chen Shao, Michael Färber, Sebastian Pütz, et al. Real-e: A foundation benchmark for advancing robust and generalizable electricity forecasting. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM)*, pages 6523–6527, 2025. doi: 10.1145/3746252.3761637.
- ENTSO-E. ENTSO-E Transparency Platform: Installed Capacity Per Production Type. <https://transparency.entsoe.eu/>, 2025. Accessed: 2025.
- Stanley Wasserman and Katherine Faust. *Social Network Analysis: Methods and Applications*. Cambridge University Press, 1994.
- Duncan J Watts and Steven H Strogatz. Collective dynamics of ‘small-world’ networks. *Nature*, 393(6684):440–442, 1998.
- Mark EJ Newman and Michelle Girvan. Finding and evaluating community structure in networks. *Physical Review E*, 69(2):026113, 2004.

Mark EJ Newman. Assortative mixing in networks. *Physical Review Letters*, 89(20):208701, 2002.